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Studierichting: Informatica

Oefeningenreeks 4A nr. 23.1

$$\int \text{Bgtan}(x) dx$$

$$\begin{cases} u = \text{Bgtan}(x) \\ dv = dx \end{cases} \Rightarrow \begin{cases} du = \frac{1}{1+x^2} dx \\ v = x \end{cases}$$

$$= x \text{Bgtan}(x) - \int \frac{x}{1+x^2} dx$$

$$\text{Stel } t = 1 + x^2 \Rightarrow dt = 2x dx$$

$$= x \text{Bgtan}(x) - \frac{1}{2} \int \frac{dt}{t}$$

$$= x \text{Bgtan}(x) - \frac{1}{2} \ln |1 + x^2| + c$$

References